

Hosting a Website on AWS EC2 (Ubuntu)

1. Objective

To launch an Ubuntu EC2 instance and host a static website using Apache Web Server.

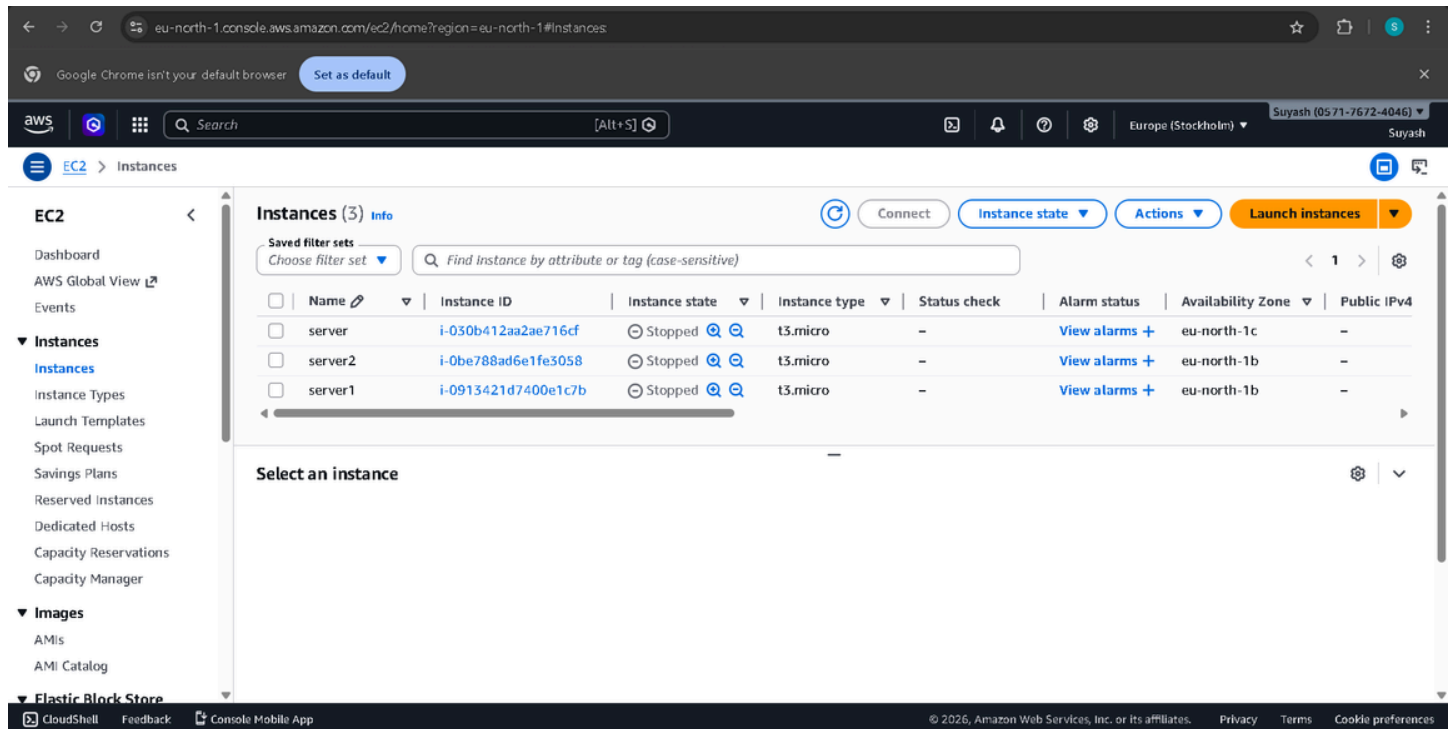
2. Requirements

- AWS Account
- Internet Connection
- Basic Linux commands knowledge

3. Steps to Create EC2 Instance

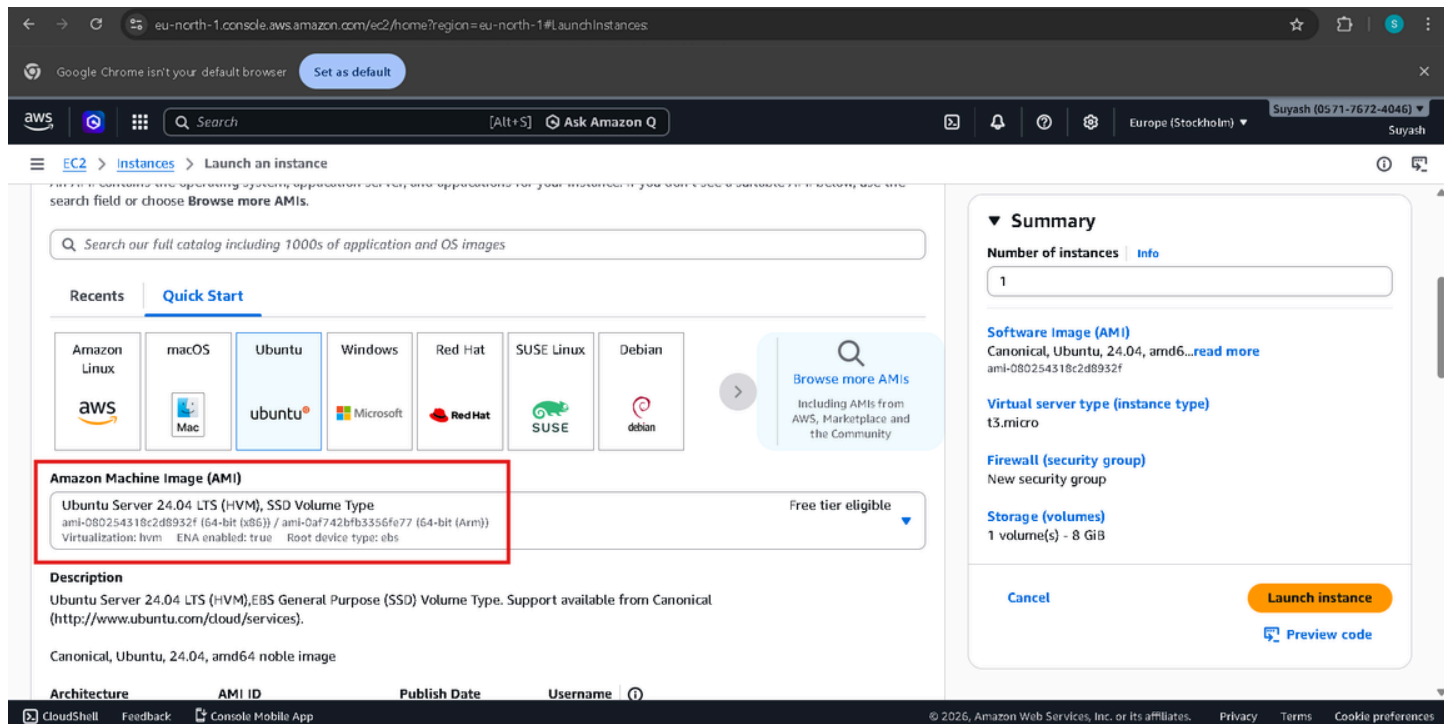
Step 1: Login to AWS

- Go to AWS Console
- Search for EC2



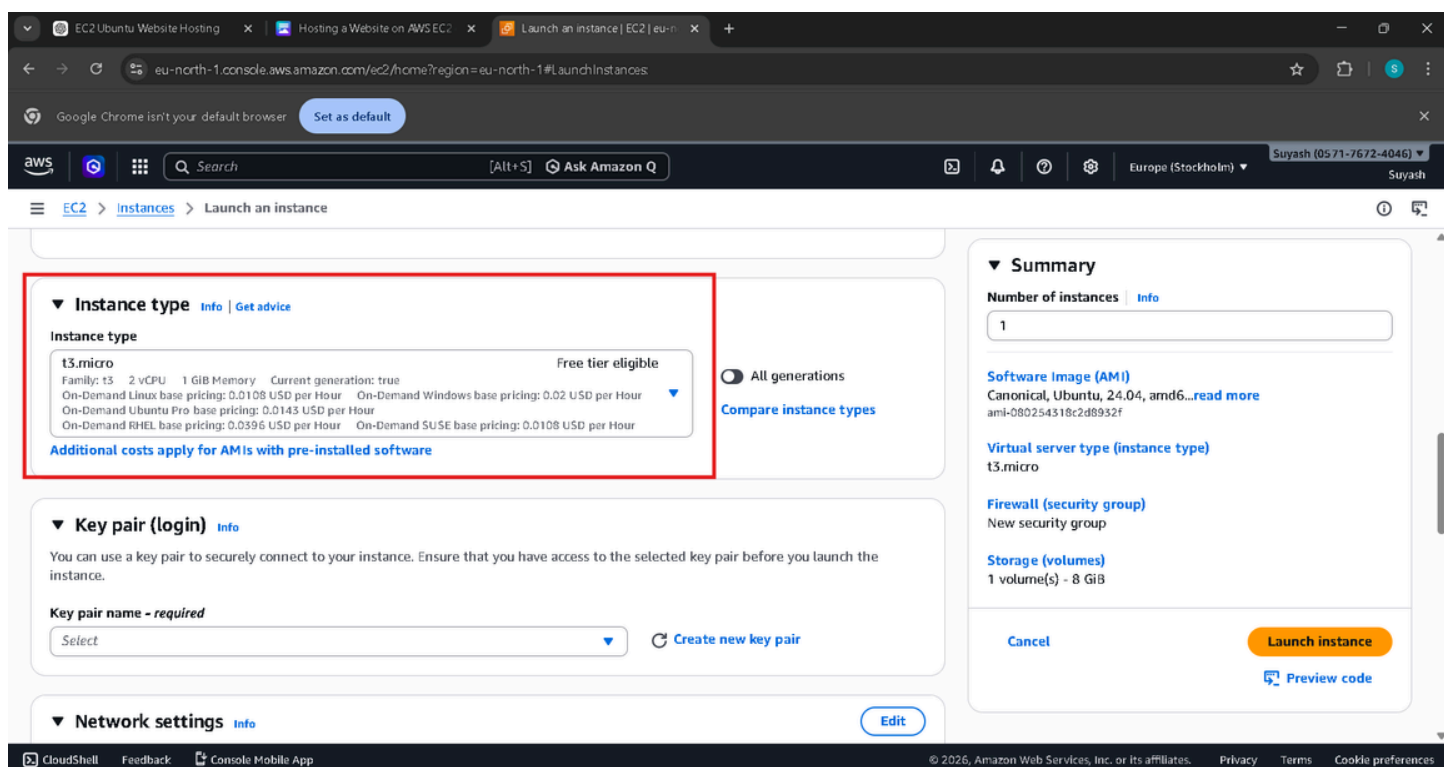
Step 2: Launch Instance

- Click Launch Instance
- Name: server



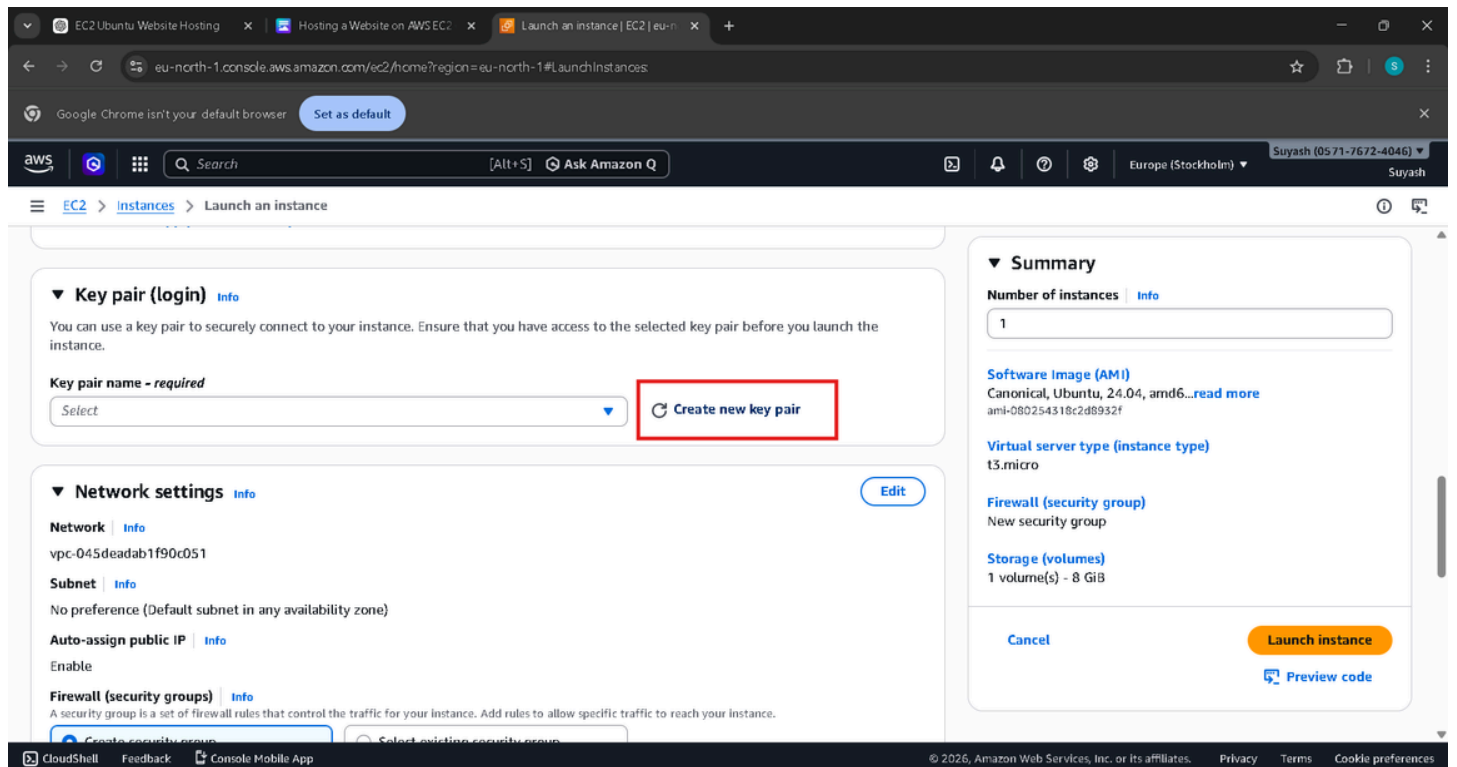
Step 4: Choose Instance Type

- Select: t2.micro (Free tier)

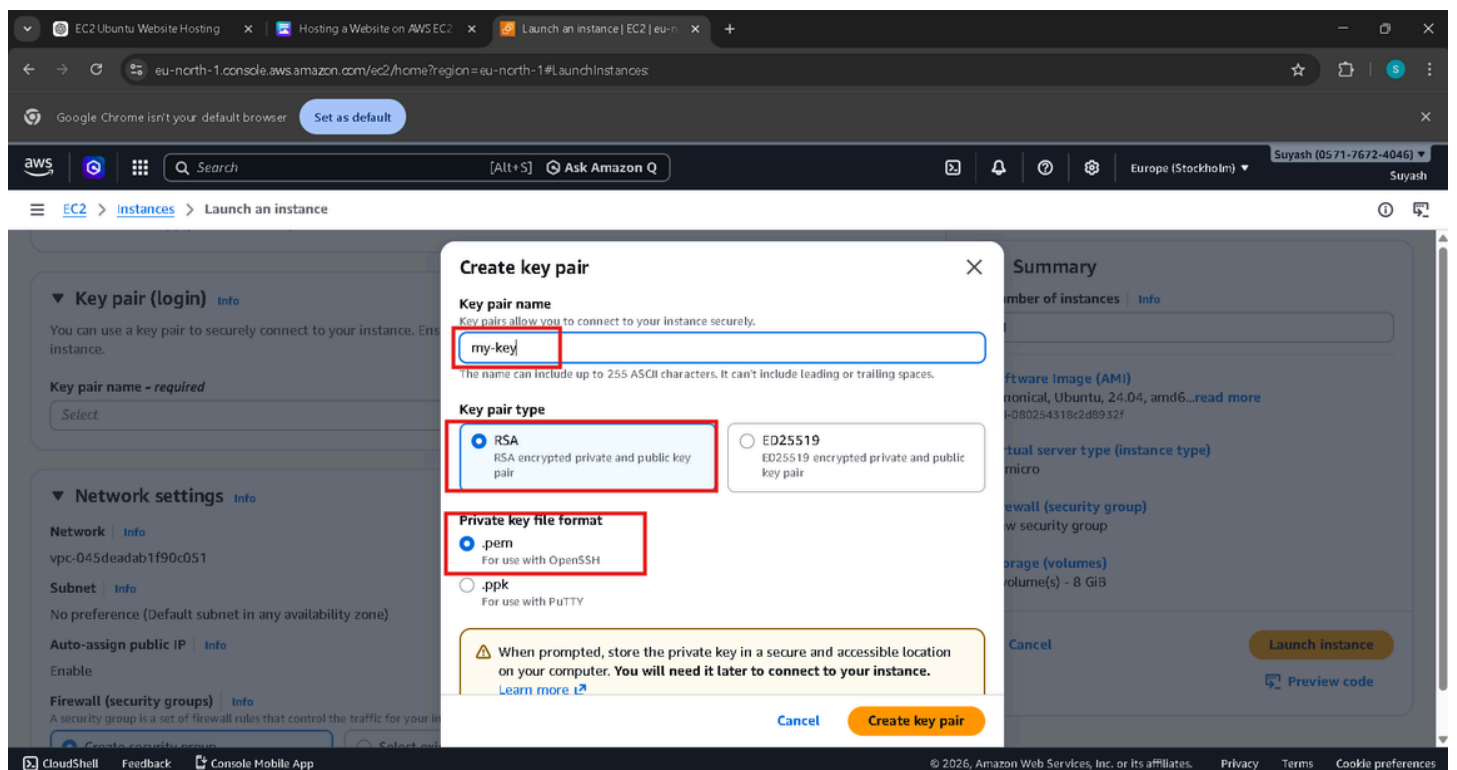


Step 5: Create Key Pair

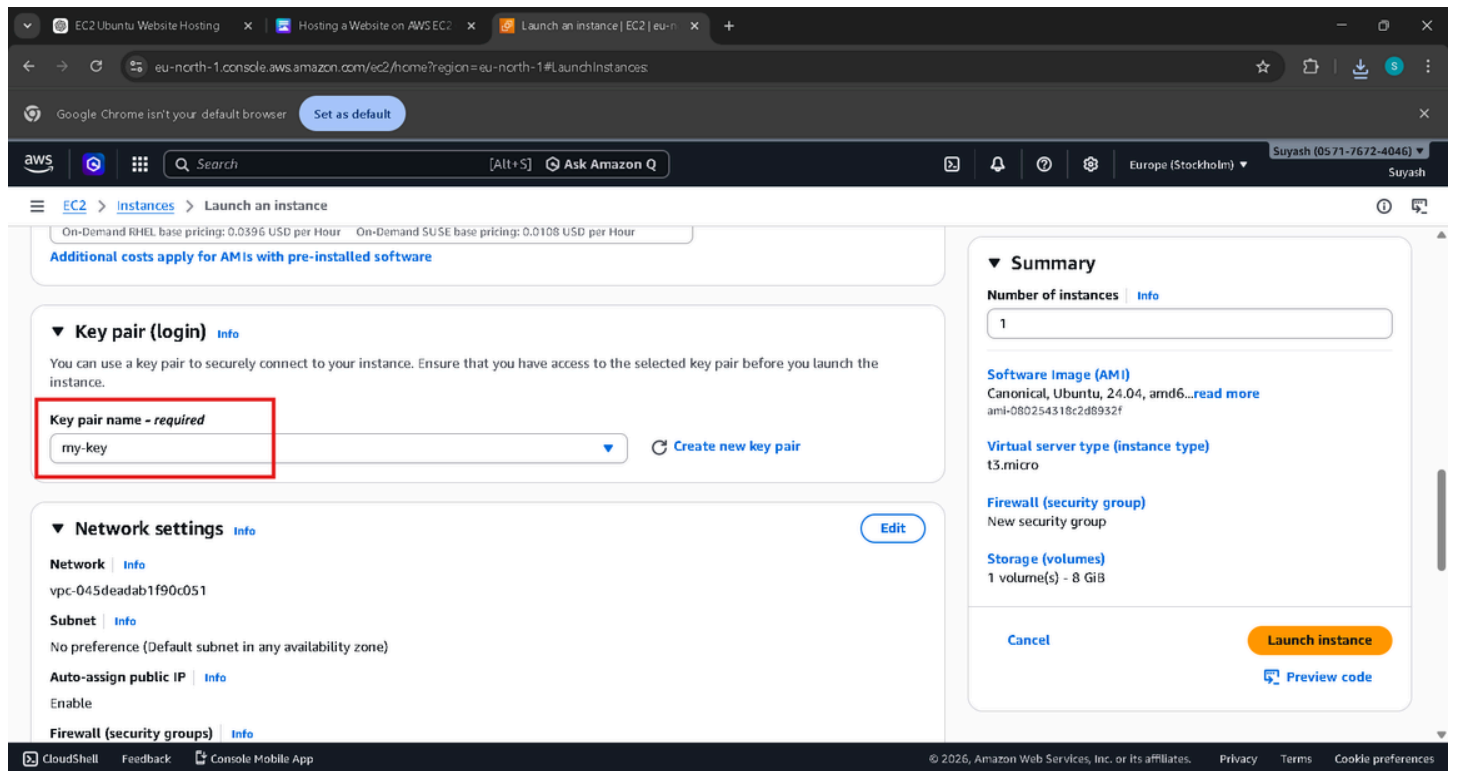
- Click Create Key Pair



- Name: my-key
- Type: .pem
- Download it

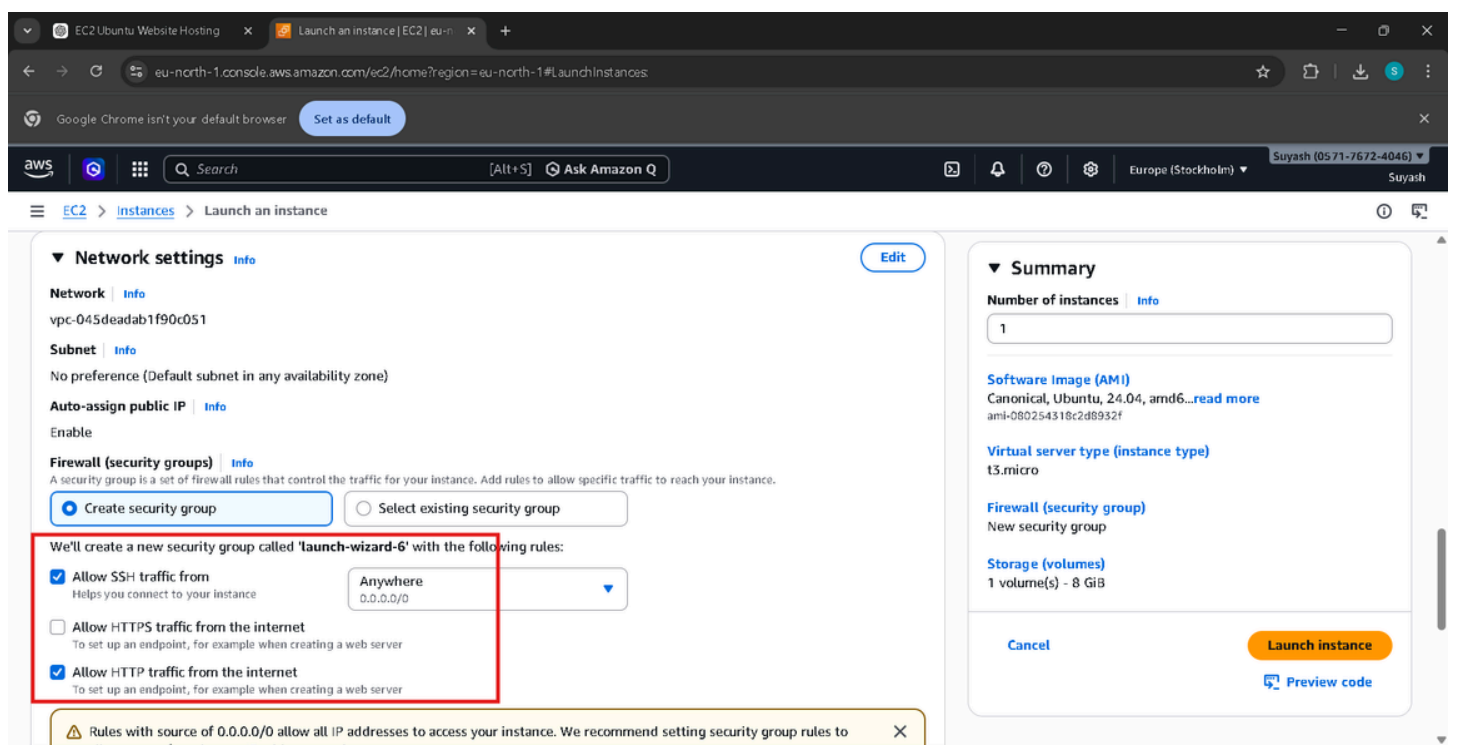


Click on Create key pair



Step 6: Configure Network

- Allow:
 - SSH (22)
 - HTTP (80)



Step 7: Launch Instance

- Click Launch

eu-north-1.console.aws.amazon.com/ec2/home?region=eu-north-1#LaunchInstances

Google Chrome isn't your default browser [Set as default](#)

aws [Search] [Alt+S] Ask Amazon Q Europe (Stockholm) Suyash (05 71-7672-4046) Suyash

EC2 > Instances > Launch an instance

Auto-assign public IP [Info](#)
Enable

Firewall (security groups) [Info](#)
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group ☐ Select existing security group

We'll create a new security group called 'launch-wizard-6' with the following rules:

☒ Allow SSH traffic from Anywhere
Helps you connect to your instance

☐ Allow HTTPS traffic from the internet
To set up an endpoint, for example when creating a web server

☒ Allow HTTP traffic from the internet
To set up an endpoint, for example when creating a web server

Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Configure storage [Info](#) [Advanced](#)

1x GiB Root volume, 3000 IOPS, Not encrypted

Summary

Number of instances [Info](#)
1

Software Image (AMI)
Canonical, Ubuntu, 24.04, amd64...[read more](#)
ami-080254318c2d8932f

Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#) [Preview code](#)

• Wait until Running

EC2 > Instances

EC2 Dashboard
AWS Global View
Events

Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts

Instances (4) [Info](#)
Last updated less than a minute ago [Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

Saved filter sets
Choose filter set [Find instance by attribute or tag \(case-sensitive\)](#)

	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4
☐	server	i-030b412aa2ae716cf	Stopped	t3.micro	-	View alarms +	eu-north-1c	-
☐	server2	i-0be788ad6e1fe3058	Stopped	t3.micro	-	View alarms +	eu-north-1b	-
☐	server1	i-0913421d7400e1c7b	Stopped	t3.micro	-	View alarms +	eu-north-1b	-
☒	server	i-0027386525bb05c5c	Running	t3.micro	Initializing	View alarms +	eu-north-1b	ec2-16-171

Select instance: server

Select an instance

Connect instance

EC2 Ubuntu Website Hosting x Instances | EC2 | eu-north-1 x +

eu-north-1.console.aws.amazon.com/ec2/home?region=eu-north-1#Instances

Google Chrome isn't your default browser [Set as default](#)

aws [Search] [Alt+S] Ask Amazon Q Europe (Stockholm) Suyash (05 71-7672-4046) Suyash

EC2 > Instances

EC2 Dashboard
AWS Global View
Events

Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Capacity Manager

Images
AMIs
AMI Catalog

Instances (1/4) [Info](#)
Last updated 1 minute ago [Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

Saved filter sets
Choose filter set [Find instance by attribute or tag \(case-sensitive\)](#)

	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4
☐	server2	i-0be788ad6e1fe3058	Stopped	t3.micro	-	View alarms +	eu-north-1b	-
☐	server1	i-0913421d7400e1c7b	Stopped	t3.micro	-	View alarms +	eu-north-1b	-
☒	server	i-0027386525bb05c5c	Running	t3.micro	Initializing	View alarms +	eu-north-1b	ec2-16-171

i-0027386525bb05c5c (server)

[Details](#) [Status and alarms](#) [Monitoring](#) [Security](#) [Networking](#) [Storage](#) [Tags](#)

Instance summary [Info](#)

Instance ID
i-0027386525bb05c5c

Public IPv4 address
16.171.20.142 [open address](#)

Private IPv4 addresses
172.31.32.155

IPv6 address
-

Instance state
Running

Public DNS
-

4. Connect to EC2 Instance

Step 8: Connect via SSH

Open terminal and Give path

C:\Users\suyas\OneDrive\Desktop\my-key.pem

```
C:\Users\suyas>C:\Users\suyas\OneDrive\Desktop\my-key.pem

C:\Users\suyas>ssh -i "C:\Users\suyas\OneDrive\Desktop\my-key.pem" ubuntu@16.171.20.142
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-1007-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Tue Mar 31 13:45:40 UTC 2026

System load:  0.02           Temperature:   -273.1 C
Usage of /:   26.3% of 6.71GB Processes:    111
Memory usage: 24%           Users logged in: 0
Swap usage:   0%            IPv4 address for ens5: 172.31.32.155

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.
```

Give axis of Ubuntu to CLI

Command -ssh -i "C:\Users\suyas\OneDrive\Desktop\my-key.pem" ubuntu@16.171.20.142

SUCCESS OUTPUT

You should see:

```
Welcome to Ubuntu...
ubuntu@ip-xxx-xxx:~$
```

```
C:\Users\suyas>C:\Users\suyas\OneDrive\Desktop\my-key.pem
C:\Users\suyas>ssh -i "C:\Users\suyas\OneDrive\Desktop\my-key.pem" ubuntu@16.171.20.142
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The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-172-31-32-155:~$ |
```

now let's host your website on EC2 (Ubuntu)

STEP 1: Update System

Run:

sudo apt update

```
ubuntu@ip-172-31-32-155: ~$ sudo apt update
Hit:1 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:4 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:5 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 Packages [15.0 MB]
Get:6 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1551 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [250 kB]
Get:8 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.5 kB]
Get:9 http://security.ubuntu.com/ubuntu noble-security/main amd64 c-n-f Metadata [10.7 kB]
Get:10 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1166 kB]
Get:11 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [225 kB]
Get:12 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [74.2 kB]
Get:13 http://security.ubuntu.com/ubuntu noble-security/universe amd64 c-n-f Metadata [22.9 kB]
Get:14 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [2701 kB]
Get:15 http://security.ubuntu.com/ubuntu noble-security/restricted Translation-en [627 kB]
Get:16 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [212 B]
Get:17 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Packages [28.8 kB]
Get:18 http://security.ubuntu.com/ubuntu noble-security/multiverse Translation-en [6980 B]
Get:19 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [212 B]
Get:20 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 c-n-f Metadata [396 B]
Get:21 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble/universe Translation-en [5982 kB]
Get:22 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 Components [3871 kB]
Get:23 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 c-n-f Metadata [301 kB]
Get:24 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64 Packages [269 kB]
Get:25 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse Translation-en [118 kB]
Get:26 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64 Components [35.0 kB]
Get:27 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64 c-n-f Metadata [8328 B]
Get:28 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1847 kB]
Get:29 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main Translation-en [339 kB]
Get:30 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [177 kB]
Get:31 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [16.9 kB]
Get:32 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1665 kB]
Get:33 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [323 kB]
Get:34 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [386 kB]
Get:35 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 c-n-f Metadata [34.2 kB]
Get:36 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [2837 kB]
Get:37 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [656 kB]
Get:38 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
```



```

Get:46 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-backports/main amd64 c-n-f Metadata [368 B]
Get:47 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-backports/universe amd64 Packages [30.7 kB]
Get:48 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-backports/universe Translation-en [18.2 kB]
Get:49 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [13.2 kB]
Get:50 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-backports/universe amd64 c-n-f Metadata [1480 B]
Get:51 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [216 B]
Get:52 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-backports/restricted amd64 c-n-f Metadata [116 B]
Get:53 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Packages [780 B]
Get:54 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-backports/multiverse Translation-en [372 B]
Get:55 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Components [212 B]
Get:56 http://eu-north-1.ec2.archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 c-n-f Metadata [116 B]
Fetched 41.1 MB in 6s (6573 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
30 packages can be upgraded. Run 'apt list --upgradable' to see them.
ubuntu@ip-172-31-32-155:~$ |

```

STEP 2: Install Apache (Web Server)

command- `sudo apt install apache2 -y`

```

ubuntu@ip-172-31-32-155:~$ sudo apt install apache2 -y
Enabling module authz_host.
Enabling module authn_core.
Enabling module auth_basic.
Enabling module access_compat.
Enabling module authn_file.
Enabling module authz_user.
Enabling module alias.
Enabling module dir.
Enabling module autoindex.
Enabling module env.
Enabling module mime.
Enabling module negotiation.
Enabling module setenvif.
Enabling module filter.
Enabling module deflate.
Enabling module status.
Enabling module reqtimeout.
Enabling conf charset.
Enabling conf localized-error-pages.
Enabling conf other-vhosts-access-log.
Enabling conf security.
Enabling conf serve-cgi-bin.
Enabling site 000-default.
Created symlink /etc/systemd/system/multi-user.target.wants/apache2.service → /usr/lib/systemd/system/apache2.service.
Created symlink /etc/systemd/system/multi-user.target.wants/apache-htcacheclean.service → /usr/lib/systemd/system/apache-htcacheclean.service.
Processing triggers for ufw (0.36.2-6) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.7) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-32-155:~$ |

```

STEP 3: Check Apache Status

Run:

`sudo systemctl status apache2`

Expected Output:

- You should see:

active (running)

```
ubuntu@ip-172-31-32-155: ~  
Running kernel seems to be up-to-date.  
No services need to be restarted.  
No containers need to be restarted.  
No user sessions are running outdated binaries.  
No VM guests are running outdated hypervisor (qemu) binaries on this host.  
ubuntu@ip-172-31-32-155:~$ sudo systemctl status apache2  
● apache2.service - The Apache HTTP Server  
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)  
   Active: active (running) since Tue 2026-03-31 14:04:23 UTC; 4min 11s ago  
     Docs: https://httpd.apache.org/docs/2.4/  
   Main PID: 2227 (apache2)  
     Tasks: 55 (limit: 1004)  
    Memory: 5.3M (peak: 5.6M)  
       CPU: 45ms  
   CGroup: /system.slice/apache2.service  
           └─2227 /usr/sbin/apache2 -k start  
             └─2230 /usr/sbin/apache2 -k start  
               └─2231 /usr/sbin/apache2 -k start  
  
Mar 31 14:04:23 ip-172-31-32-155 systemd[1]: Starting apache2.service - The Apache HTTP Server...  
Mar 31 14:04:23 ip-172-31-32-155 systemd[1]: Started apache2.service - The Apache HTTP Server...  
lines 1-15/15 (END)  
● apache2.service - The Apache HTTP Server  
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)  
   Active: active (running) since Tue 2026-03-31 14:04:23 UTC; 4min 11s ago  
     Docs: https://httpd.apache.org/docs/2.4/  
   Main PID: 2227 (apache2)  
     Tasks: 55 (limit: 1004)  
    Memory: 5.3M (peak: 5.6M)  
       CPU: 45ms  
   CGroup: /system.slice/apache2.service  
           └─2227 /usr/sbin/apache2 -k start  
             └─2230 /usr/sbin/apache2 -k start  
               └─2231 /usr/sbin/apache2 -k start  
  
Mar 31 14:04:23 ip-172-31-32-155 systemd[1]: Starting apache2.service - The Apache HTTP Server...
```

STEP 3: Open Default Website

Open browser and enter:

<http://16.171.20.142>

Output:

Apache default page (“It works!”)



Go to web folder

cd /var/www/htm

```
ubuntu@ip-172-31-32-155: /v...
C:\Users\suyas>cd /var/www/html
The system cannot find the path specified.

C:\Users\suyas>ssh -i "C:\Users\suyas\OneDrive\Desktop\my-key.pem" ubuntu@16.171.20.142
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-1007-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Tue Mar 31 16:51:16 UTC 2026

System load:  0.0          Temperature:   -273.1 C
Usage of /:   30.5% of 6.71GB Processes:      113
Memory usage: 28%         Users logged in: 0
Swap usage:   0%          IPv4 address for ens5: 172.31.32.155

 * Ubuntu Pro delivers the most comprehensive open source security and
   compliance features.
   https://ubuntu.com/aws/pro

Expanded Security Maintenance for Applications is not enabled.

34 updates can be applied immediately.
33 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Tue Mar 31 14:28:25 2026 from 166.192.132.71
ubuntu@ip-172-31-32-155:~$
ubuntu@ip-172-31-32-155:~$ cd /var/www/html

ubuntu@ip-172-31-32-155:/var/www/html$
ubuntu@ip-172-31-32-155:/var/www/html$
ubuntu@ip-172-31-32-155:/var/www/html$ cd /var/www/html
ubuntu@ip-172-31-32-155:/var/www/html$ |
```

Remove default page

sudo rm index.html

Create your website file

sudo nano index.html

```
ubuntu@ip-172-31-32-155: /v...
GNU nano 7.2 index.html
<h1>My First AWS Website 🚀</h1>
<p>Hosted using EC2 and Apache</p>
```

Add your content

Paste this:

```
<h1>My First AWS Website </h1>
<p>Hosted using EC2 and Apache</p>
```

```
ubuntu@ip-172-31-32-155:/var/www/html$
ubuntu@ip-172-31-32-155:/var/www/html$
ubuntu@ip-172-31-32-155:/var/www/html$ cd /var/www/html
ubuntu@ip-172-31-32-155:/var/www/html$ cat html
cat: html: No such file or directory
ubuntu@ip-172-31-32-155:/var/www/html$ cat index.html
<h1>My First AWS Website 🚀</h1>
<p>Hosted using EC2 and Apache</p>
ubuntu@ip-172-31-32-155:/var/www/html$ |
```

Save file

- Press CTRL + X
- Press Y
- Press Enter

Open in browser

Go to:

<http://16.171.20.142>



My First AWS Website

Hosted using EC2 and Apache

